Ryan D. Heberlig

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EDUCATION

Northeastern University, Boston MA – *B.S. in Mechanical Engineering* **Expected Graduation 2028**

- 3.92 GPA; Dean's List
- Related Coursework: Mechanical Computation and Design, Dynamics, Fluids, Thermodynamics

SKILLS & CERTIFICATIONS

Technical Skills: Fixture Design, GD&T, DFM & DFA, Rapid Prototyping, FEA
Software & Tools: Creo, SolidWorks, Onshape, Ansys, MATLAB, Arduino
Manufacturing: Manual Mill & Lathe, Laser Cutting, SLA & FDM 3D Printing, Soldering
Certifications: Lean Six Sigma Green Belt, CSWA, Tec-Ease GD&T Fundamentals

WORK EXPERIENCE

Zaiput Flow Technologies, Waltham MA – *Mechanical Engineering Co-Op* **Incoming: Spring 2026**

- Incoming R&D engineering co-op supporting SolidWorks design, fabrication, and testing of flow chemistry equipment for pharmaceutical manufacturing

Amphenol PCD, Beverly MA – *Manufacturing Engineering Co-Op* **Jan 2026-Jun 2026**

- Designed 18 manufacturing fixtures & tools with Creo drawings per ASME Y14.5, including a pneumatic diode lead-bending fixture that cut process time by **89%**.
- Led integration of laser engraving to replace ink marking across 7 product lines, reducing part-marking time by up to **6x**.
- Developed a raceway power distribution system for low-pressure molding, isolating failures to quick-swap parts, eliminating downtime and raising concurrent molding capacity by **25%**.
- Developed and standardized production processes, writing manufacturing work instructions, and leading training sessions for operators and engineers.
- Repaired industrial machinery, diagnosing equipment failures using root cause analysis.

PROJECTS

Generate Product Development Studio – *Hardware Engineer* **Sep 2026-Present**

- Designing a single-motor rotary indexing mechanism for an automated pill dispenser's travel unit.
- Evaluated 6 drive architectures via torque and FOS analysis; selected a self-locking worm motor with 10× torque margin and drift-free hall-effect indexing.

Forge Product Lab – *Hardware Team Member* **Feb 2026-Apr 2026**

- Created a braille E-reader alongside a team of 6 able to translate PDF text into braille.
- Designed a six-pin actuation mechanism in Onshape to drive standard **6 × 4 mm** refreshable braille cells, independently driving pins **1.5 mm** in diameter.
- Organized ME system meetings and led design reviews alongside EE Lead.
- Fabricated FDM and SLA 3D-printed prototypes integrated with N20 motors & other electronics.

Northeastern Electric Racing – *Member* **Sep 2024-Dec 2025**

- Designed the front anti-roll bar for the team's race car, including geometry selection, material choice, stress analysis and stiffness calculations to optimize handling and cornering performance.
- Optimized parts using FEA, reducing weight by up to **60%** while maintaining minimum FOS.

Punching StrengthTester – *Cornerstone of Engineering I* **Oct 2024-Dec 2024**

- Co-developed strength tester rated for **450+ N** loads, laser cut target and electronics housing.
- Programmed an accelerometer in C++ on Arduino to capture motion data and convert it into a real-time scoring system

Special Interests: Product Development, Volleyball, Reading, Drawing, Hiking